| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 7(a)(ii) | 2.4[0] | **4** | **B3** for answer in range 2.396… to 2.40…<br>OR<br>**M3** for *their* $\frac{16\pi}{3} + \pi \times 2^2 \times 5.2 + \frac{1}{3}\pi \times 2^2 \times h = \frac{88\pi}{3}$ oe<br>or **M2** for $\frac{88\pi}{3}$ – *their* $\frac{16\pi}{3} - \pi \times 2^2 \times 5.2$ oe<br>or **M1** for $\pi \times 2^2 \times 5.2$ oe<br>or $\frac{1}{3}\pi \times 2^2 \times h$ oe soi |
| 7(a)(iii) | 1 hour 38 min or 1 hour 37.8 min to 1 hour 37.9… min | **3** | **B2** for 1.63[2…] or 98 [mins] or 97.8 to 97.9… ]<br>or **M1** for $\frac{\frac{88\pi}{3} \times 620}{35000}$ [× 60] oe |
| 7(b) | 8.5[0] or 8.496 to 8.497 | **4** | **M3** for [$r$= ] $\sqrt{\frac{65}{\frac{140}{360}\pi - \frac{1}{2}\sin 140}}$ oe<br>or **M2** for $\frac{140}{360}\pi \times r^2 - \frac{1}{2}r^2 \times \sin 140$ [=65] oe<br><br>or **M1** for either area expression seen |
| 8(a)(i) | $\frac{12}{x}$ or $12 \div x$ final answer | **1** | |
| 8(a)(ii) | $\frac{12}{x-4}$ – *their* $\frac{12}{x} = 1.5$oe | **M1** | Accept 3 or more term equivalents |
| | $12x - 12(x-4) = 1.5x(x-4)$<br>or<br>$\frac{12x - 12(x-4)}{x(x-4)}$ [= 1.5] | **M1** | Correctly clearing fractions, or correctly collecting into a 'single fraction'<br>FT *their* expression dep on two fractions both with algebraic denominators |
| | $12x - 12x + 48 = 1.5x^2 - 6x$ | **M1** | Correctly multiplying *their* two sets of brackets<br>FT *their* expression dep on two fractions both with algebraic denominators or first M1 given |
| | [$1.5x^2 - 6x - 48 = 0$ ]<br><br>$x^2 - 4x - 32 = 0$ | **A1** | One further step either 3 term equation or division throughout by 1.5 leading to solution<br><br>With no errors or omissions seen, dep on **M3** |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 8(a)(iii) | $(x + 4)(x - 8)$ | **M2** | **M1** for $(x + a)(x + b)$<br>where $ab = -32$ or $a + b = -4$<br>or for $x(x + 4) - 8(x + 4)$<br>or $x(x - 8) + 4(x - 8)$ |
| | –4 and 8 | **B1** | |
| 8(a)(iv) | 3 | **2** | **FT** $\frac{12}{their\ 8 - 4}$<br>**M1** for $\frac{12}{their\ 8 - 4}$ or $\frac{12}{their\ 8} + 1.5$ oe<br>or for answer $\frac{12}{their\ 8}$ |
| 8(b) | 69.6 | **3** | **M2** for $\frac{430 \text{ to } 440}{6 + 0.25}$ or $\frac{440 - 5}{6 \text{ to } 6.5}$ oe<br>or **M1** for 440 + 5 oe or 440 – 5 oe<br>or 6 + 0.25 oe or 6 – 0.25 oe seen |
| 9(a)(i) | (3, 1) | **1** | |
| 9(a)(ii) | $\begin{pmatrix} -10 \\ 15 \end{pmatrix}$ | **1** | |
| 9(a)(iii) | 3.61 or 3.605 to 3.606 | **2** | **M1** for $(-2)^2 + 3^2$ oe |
| 9(b)(i)(a) | $\frac{1}{2}\mathbf{c}$ | **1** | |
| 9(b)(i)(b) | $\mathbf{a} + \frac{1}{2}\mathbf{c}$ oe | **1** | FT **a** + *their* (b)(i)(a) |